

Agustín Buffa

Aerospace Engineering | Aerodynamics | Flight Simulation | Small Satellites

EU citizen (Italy) · Available to relocate immediately | agusbuffa.ab@gmail.com

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Profile

Aeronautical Engineering student (UTN, MSc-equivalent, GPA 9.1/10; degree expected December 2026) with research experience in morphing wing aerodynamics, aerodynamic modeling for flight simulation, CFD, and small satellite systems. Italian and Argentine citizen (no work permit required within the EU).

Education

UTN, Facultad Regional Haedo, Aeronautical Engineering – Argentina 2020 – 2026

- GPA: 9.1/10. Thesis: Lifting-Line Theory for morphing wing aerodynamic optimisation.

Technische Universität Braunschweig, Aerospace Engineering (exchange) – Germany 2025 – 2026

- DAAD KOSPIE Scholarship (selected among 150 applicants); Configuration Aerodynamics, MDO.

Experience

Intern, Unmanned Aircraft Systems Department, DLR, Institut für Flugsystemtechnik (IFS) – Braunschweig, Germany Mar 2026 – Jul 2026

Project morphAIR: aerodynamic modeling of fixed-wing UAVs with adaptive wings (PROTEUS platform).

- Evaluated AVL, VSPAERO, and XFLR5 (VLM) against lifting-line theory (LLT) for flight simulation.
- Developed an LLT-based optimization method to obtain drag-minimizing spanwise deflection distributions.
- Conceived an aerodynamic FMU and a lookup table structure for adaptive wing configurations.

Research Fellow, Laboratory for Aerodynamics & Fluid Mechanics, UTN Facultad Regional Haedo – Argentina Mar 2023 – Jul 2026

- CFD studies of morphing wing airfoils with RANS (k-omega SST, Spalart-Allmaras) and γ-Reθ models.
- Qualitative wind tunnel tests using oilflow and smoke visualization.

Research Fellow, Aerospace Technology Group, UTN Facultad Regional Haedo – Argentina Mar 2022 – Dec 2023

- Designed and built a melting wire hold, restrain and release mechanism for a 3U CubeSat.
- Carried out the preliminary design of the attitude determination and control subsystem (ADCS) for a 3U CubeSat.

Intern, LIA Aerospace – Buenos Aires, Argentina Sep 2024 – Mar 2025

- Assessed the undocumented oxidizer production system and prepared its piping diagrams.

Publications

Lifting Line Theory for Morphing Wing Applications Mar 2026

Agustín Buffa, Dr. Ing. Pablo A. Carón, Mg. Ing. Pablo N. Zitelli

zenodo.org/records/21998900 (Preprint, submitted to CEAS Aeronautical Journal)

Skills

Programming: Python, MATLAB, Octave, Linux, Git

Aerodynamics and CFD: AVL, XFOIL, XFLR5, VSPAERO, LLT and VLM methods, OpenFOAM, ANSA

Simulation, CAD and testing: FMI/FMU, SolidWorks, OpenVSP, 3D printing, wind tunnel testing

Languages: Spanish (Native), English (C1), French (B2), German (B2), Italian (A1)